Evaluation Report

Scope and evidence

This report summarizes measured results already present in evaluation/model_metrics.json. The file identifies a 100,000-row dataset, 70,000 training rows, 15,000 validation rows, and 15,000 held-out test rows. The saved report timestamp is 2026-10-05T20:28:32Z. These are recorded training-run metrics, not a fresh run performed during this repository audit.

Held-out model results

Target	Model	Test metrics
Production impact (%)	Random Forest regressor	MAE 7.99 percentage points; RMSE 9.92; R^2 0.684
Revenue loss (USD)	Random Forest regressor	MAE $1,774,595; *RMSE* 4,419,908; R^2 0.374
Full recovery time (days)	Random Forest regressor	MAE 31.74 days; RMSE 45.21; R^2 0.369
Permanent supplier change	Random Forest classifier	Accuracy 0.640; precision 0.332; recall 0.507; F1 0.401; ROC-AUC 0.635

The training script selects between Random Forest and Extra Trees by validation MAE for regression or F1 for classification. It excludes outcome and post-response fields from initial-event features to limit target/future leakage. These metrics alone do not establish operational utility. In particular, revenue and recovery errors are large relative to many likely use cases, and the supplier-change classifier has modest precision and F1.

RAG and answer quality

Faithfulness, answer relevancy, contextual relevancy, contextual precision, recommendation quality, and explanation quality are Not evaluated in the checked-in run evidence. evaluation/run_evaluation.py implements DeepEval metrics over saved analyses, but requires an LLM provider/judge and database records with retrieval evidence. This environment has no MySQL service, DeepEval install, configured judge, or saved eligible analysis examples. No RAGAS or DeepEval score is claimed.

Other validation dimensions

Dimension	Result
Retrieval relevance	Not evaluated against a labeled query/case set. Runtime hybrid scores are ranking signals, not relevance judgments.
Completeness	Not evaluated with human ratings or a reference answer set.
Factual consistency / groundedness	Not evaluated by an independent judge. Workflow validation only checks strategy support and evidence count.
Response quality	Not evaluated by human reviewers or an LLM judge.
Validation accuracy	Model classification accuracy is 0.640 for permanent supplier change; this does not represent general API or recommendation validation accuracy.

Sample test cases

Automated API and retriever helper test cases live in tests/. Their status is the result of the repository test run, not an LLM quality evaluation. Expected edge behavior includes HTTP 422 on malformed/empty search queries, HTTP 404 for unknown records, HTTP 503 when retrieval artifacts are missing, top-K truncation, and an empty result list when there are no matches.

Reproduction

Install the root requirements, configure MySQL, then run python scripts/train_models.py to produce a new timestamped model report. For answer quality, configure LLM_PROVIDER and LLM_MODEL, save analyses with retrieved cases, then run python evaluation/run_evaluation.py. Record the judge model and output report; do not treat absent scores as zero.